

User Manual

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1 Introduction

This is the user manual for the Converter program. Its purpose is to convert the models present in the Caldera map data set from USD format to OBJ format.

To work properly, the program requires access to the data set with an intact folder hierarchy. It also requires an operational OpenUSD framework (for more information, refer to the README file or the Converter gitlab repository).



Figure 1: The base view of the Converter program with 4 highlighted and numbered UI windows

2 UI navigation

The UI is organized into 4 windows:

- Converter Tools (1.)
- Included Paths (2.)
- Export Options (3.)
- Renderer Controls (4.)

2.1 Converter Tools

This window offers the option to switch between two rendering modes ("full faces" and "wireframe"). It also shows the number of faces (triangles) loaded and rendered in the scene. The following are then 3 sliders that modify the camera speed control

2.2 Included Paths

The window for path management. Each path included loads and renders one model. Each path needs to be added through the "Add Path" button. Multiple paths can be added and loaded together (models may overlap if model transformations do not match, however; Multi-path render is only recommended when loading a single tile made of multiple layers among multiple paths) An added path can be selected and deleted via the "Remove Path" button. When all requested paths are added, "Import and Render" will load models on added paths and render them.

2.3 Export Options

The window with the option to export the reduced model. This option is recommended when exporting only a part of the model (either through a bounding box select or through random sampling), as it ignores redundant vertices and therefore lowers the size of the output file. The "Export" button will open the window to select the name and location where the model should be exported.

2.4 Rendered Controls

This window contains 6 sliders that adjust what part of the model is selected for export. The sliders are separated by axes into 3 groups. Each group contains 2 sliders, one that modifies the size of the box along the given axis ('x' size slider) and one that moves the box along the given axis ('x' offset slider). The "Clear View" removes all loaded and rendered models.